

Comparative Economics of Main Season and Spring Rice Production in Kanchanpur District, Nepal

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Abstract: Rice is the major staple food crop in Nepal which has been growing in two different seasons mainly June/July – October/November as main season rice (MSR) whereas during February/March – June/July as spring rice (SR). A comparative economic analysis of MSR and SR production in Kanchanpur district Nepal was conducted during March to July 2023. A total of 120 rice-growing farmers including 60 MSR and 60 SR growers were randomly selected from the available grower list to collect primary information by using pre-tested interview schedules. Descriptive statistics, independent sample t-tests, and the Cobb-Douglas production function were employed for the data analysis. The study finding revealed that the average landholding of MSR growers was 0.65 ha/hh, whereas the same for SR growers, was 0.55 ha/hh. The average area for MSR and SR were at 0.53 ha and 0.29 ha, respectively. The average productivity of the SR (3.69 t/ha) in the study area was found to be significantly higher than the MSR (4.61 t/ha). Despite the higher cost of production for SR (Rs. 80309.95/ha) than MSR (Rs. 75048.45/ha), the Benefit-Cost ratio for SR was found to be higher (1.61) than that of MSR (1.38). The analysis of coefficients indicated a decreasing for main season rice and increasing return to scale for spring rice, with sums of coefficients measuring 0.74 and 1.24 respectively. This implies that a 100% increase in expenditure on the variables outlined in the model for MSR and SR led to a 74% and 95% increase in the gross return of rice production. These results underscore the potential advantages of promoting improved seed varieties and adopting commercial farming practices in the spring rice cultivation area. The findings indicated the need to replace manual labor with machinery to reduce costs and increase rice cultivation profits.

Keywords: BC ratio, Cobb-Douglas, Economics, Gross margin, Spring

I. INTRODUCTION

Agriculture has been considered as a vital sector of a country's economy which contributes 25.83% of GDP and provides employment to 60.4% of the total population (AITC, 2022) along with its magnificent contribution to food security, employment generation and poverty reduction.

Among different crops being grown at commercial scale, rice (*Oryza sativa*) is the major staple food crop cultivated across a diverse array of agroecological zones encompassing varying climates, altitudes, and topography, spanning from 60 meters in the Terai to 3050 meters above sea level in Chumchure and Jumla (Devkota *et al.*, 2022). Rice stands as the foremost crop, serving as a fundamental cornerstone of livelihood and income for over two-thirds of farming households, while also being intricately woven into the cultural fabric of the nation (Poudel *et al.*, 2021). Rice constitutes 20% of Nepal's agricultural gross domestic product, while also playing a pivotal role by fulfilling approximately 50% of the total caloric requirements of the Nepalese

population (NARC, 2016). Nepal's population, growing at an annual rate of 1.25%, necessitates addressing rising demand, leading the country to import approximately 215 thousand tons of rice annually. National statistics data reveals a 1.2% decrease in rice production between 2015 and 2016 (CBS, 2017).

Rice is grown in two seasons in the district namely spring rice (SR) is planted in the month of February/March and harvested during June/July while main season rice (MSR) is planted in the month of June/July and harvested during October/November. In Nepal, SR is also commonly known as Chaite rice which constitutes 8% of the overall rice supply (MoALD, 2019). When compared to MSR, SR emerges as a short-duration crop, effectively harnessing the potential of the spring season for cultivation. Typically, photoperiod-insensitive rice varieties are cultivated during this period. The harvest of spring rice occurs within three to four months, aligning with the advent of the rainy season when planted during spring. SR generally commands a higher price than MSR (Airee *et al.*, 2020). In regions like Terai, inner Terai, river basins, and similar climatic zones, about 11 types of spring paddy, including CH45, Bindeshwori, Chaite-2, Chaite-5, Hardinath-1, and Chaite-6, have been introduced. Hardinath-1 is prominent in most Chaite rice cultivation areas, while Chaite-2 prevails in the eastern Terai region (Khatriwada & Upreti, 2008). But in case of main season rice, the popular varieties of main season rice grown in Kanchanpur district are Sarju-52, Radha-4, Bahuguni, Hardinath, Sawa mansuli, Sabitri, etc (Airee *et al.*, 2020).

Kanchanpur district is one of the major rice-producing districts, well known for employing the rice-wheat cropping system, dedicates 48,749 hectares to rice cultivation, yielding 187,274 metric tons with a productivity rate of 3.84 metric tons per hectare. (MoALD, 2022). SR is cultivated over 60 hectares with a productivity of 5.14 mt/ha, while the MSR covers 48,689 hectares with yield of 3.84 mt/ha (MoALD, 2022). Farmers in Kanchanpur mostly rely on subsistence and conventional farming methods due to limited skills and technology adoption. Post-harvest losses for rice, amounting to 25-30% of the seed and grain value, have been documented (Airee *et al.*, 2020). The increasing labor shortage for farm activities with increased rate of migration of youths to India and other countries for employment opportunity has increased the labor cost for rice cultivation and reduced the cultivated area (Dhakal *et al.*, 2019). Integrating machinery like power tillers, tractors, and threshers could reduce costs, streamline cultivation, and save time. (Pokhrel *et al.*, 2021).

This research aims to bridge this critical gap by comprehensively analysing the production economics, resource utilization patterns, and constraints experienced by main season rice and spring rice farmers in Kanchanpur district. This study aims to offer insights for informing policies and interventions, enhancing rice production practices, ensuring economic viability, and contributing to long-term food security goals.

II. MATERIALS AND METHODS

A. Selection of Study Site

This study was conducted in five LG areas namely Bheemdatt, Bedkot, Mahakali, Punarbas municipalities, and Beldandi rural municipality of Kanchanpur district of Sudurpashim province. The area was purposively selected based on production record of Prime Minister Agriculture Modernization Project (PMAMP), and availability of both MSR and SR producing farmers in the district. Kanchanpur district covers an area of 1610 sq. km, with altitude ranging from 176 to 1528 meters above sea level (masl). It extended from 28°-38" to 29°-28" northern latitude and 80°-33" eastern longitude (Airee *et al.*, 2020). The dominant farming system is Rice-Wheat-Maize where farmers also cultivate vegetables and legumes during winter and summer seasons. Majority of the agriculture production area have irrigation facility for rice cultivation during rainy season whereas Rice-Wheat/Vegetable-Rice is also found in the district where irrigation facility is available during spring season.

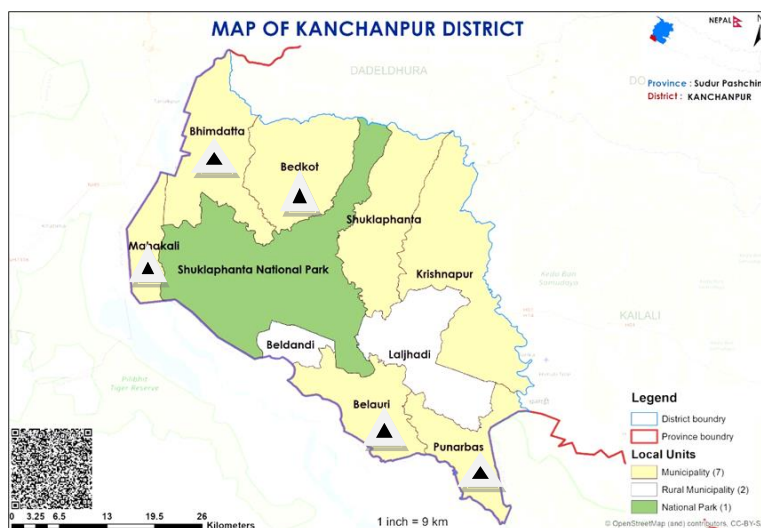


Fig. 1 Map of study area

B. Source of Data

Primary data was collected from rice farmers within the study area. The household survey was conducted using a face-to-face technique, employing a pre-tested semi-structured interview schedule. Two focus group discussions (FGD) were conducted in the study area to identify the major production constraints faced by both categories of the farmers. In addition to household survey and FGD, Key informants' interviews (KII) were conducted with agricultural officers of municipality, Agriculture Knowledge Centre (AKC) and Prime Minister Agriculture Modernization Project (PMAMP). This study also employed secondary data published from Ministry of Agriculture and Livestock Development (MoALD), Nepal Agricultural and Research Council (NARC), Agriculture Knowledge Centre (AKC) and Prime Minister Agriculture Modernization Project (PMAMP) as well as peer reviewed journal articles published in different journals and proceedings.

C. Population and Sample

A total of 120 rice-growing farmers (60 SR and 60 MSR) from the study area were interviewed to collect required data to elicit demographic, educational, socio-cultural, behavioural, economic, and other information related to rice production.

D. Data Analysis

The data collected from the survey was coded and edited, and local units of measurements were standardized into scientific units. All significant primary data collected from households were entered into MS-Excel and the Statistical Package for Social Science (SPSS 26) program for analysis. The data was analysed using descriptive statistics, mean comparisons, frequency distributions, independent sample t-tests, and other relevant methods. The findings were presented through tables, bar diagrams, pie charts, and other appropriate graphical representations.

1) *Costs of Production*: The total cost of production encompassed both fixed and variable costs. The analysis of the cost of production involved calculating the various components of variable costs. Cost of cultivation is the sum of variable costs and fixed costs expressed on a per hectare basis (Acharyaa et al., 2021).

$$\text{Total cost (NRs.)} = \text{Total fixed cost} + \text{Total variable cost}$$

where,

Total variable cost = Sum of cost for (human labor + seed + FYM and chemical fertilizers and pesticides + tillage cost + irrigation cost + harvesting + threshing + others cost)

Total fixed cost (NRs.) = Land tax + Water tax + Depreciation + Repair and maintenance cost

2) *Gross Return*: The gross return will encompass the earnings from grains as well as from byproducts such as straw (Dhakal *et al.*, 2019). It will be computed as follows:

Gross Return = [(price of rice grains × total amount of rice grains) + (price of rice byproduct × total amount of rice byproduct)]

3) *Gross Margin*: Distinguishing from the gross return, the gross margin will be determined by deducting the total cost of cultivation from the gross return. Conducting a gross margin analysis offers a straightforward and efficient approach to assess various enterprises (Sapkota *et al.*, 2018).

Gross margin = Gross return – Total variable cost

4) *Gross Profit*: Gross profit, also known as net profit, is the difference between the gross margin and total fixed costs (Acharyaa *et al.*, 2021). The formula to calculate net profit is as follows:

Gross profit = Gross margin – Total fixed costs

5) *Benefit Cost Ratio*: The benefit-cost ratio serves as a crucial indicator for evaluating the economic performance of any entity, including the agricultural sector, in a swift and straightforward manner. This ratio is calculated by dividing the gross return by the total cost (Dhital *et al.*, 2023).

Benefit-cost ratio = (Gross return)/ (Total costs)

6) *Cobb-Douglas Production Function*: The Cobb-Douglas production function is extensively employed to conduct various research studies in agriculture, effectively representing the correlation between output and inputs (Dahal & Rijal, 2019). It was utilized to estimate the contributions of three independent variables in elucidating the variations in the dependent variable of rice production. The ensuing form of the Cobb-Douglas production function was employed to ascertain the impact of independent variables on output (Subedi *et al.*, 2020).

$$Y = aX_1^{b_1} X_2^{b_2} X_3^{b_3} X_4^{b_4} X_5^{b_5} e^{\mu}$$

The equation mentioned above will be converted into a log-linear form as depicted below:

$$\ln Y = \ln a + b_1 \ln X_1 + b_2 \ln X_2 + b_3 \ln X_3 + b_4 \ln X_4 + b_5 \ln X_5 + b_6 \ln X_6 + b_7 \ln X_7 + \mu$$

where,

Y = Gross returns from rice cultivation (NRs/ ha)

X₁ = Seed cost (NRs/ha)

X₂ = Labor cost (NRs/ha)

X₃ = Fertilizer cost (NRs/ha)

X₄ = Irrigation cost (NRs/ha)

X₅ = Tillage cost (NRs/ha)

X₆ = Harvesting cost (NRs/ha)

X₇ = Threshing cost (NRs/ha)

μ = Random disturbance term or error term

a = Intercept or constant term

e = Base of natural logarithm

ln = Natural logarithm

b₁, b₂, b₃, b₄, b₅, b₆ and b₇ = Coefficient of respective variables

The explanatory variables for this study were chosen based on their potential to influence yield, as outlined in prior literature.

7) *Returns to Scale*: Returns to scale pertain to the quantitative alteration in production that occurs when all inputs or factors of production are simultaneously adjusted in the same proportion. The estimation of rice production's returns to scale was achieved by aggregating the coefficients of independent variables, denoted as b₁ + b₂ + ... + b_n, from the Cobb-Douglas production function model.

If b₁+b₂+...b_n = 1, indicates a constant return to scale.

If b₁+b₂+...b_n <1, indicates a decreasing return to scale.

If b₁+b₂+...b_n >1, indicates an increasing return to scale.

8) *Indexing*: Qualitative data were utilized in the formulation of the index. The index of importance was determined using the following formula:

$$I_{imp} = \sum (Si Fi) / (N)$$

where,

I_{imp} = Index of importance

Σ = Summation

$S_i = I^{\text{th}}$ scale value

 F_i = Frequency of i^{th} importance given by respondents

 N = Total number of respondents

III. RESULTS AND DISCUSSION

In this section, the findings on the socio-demographic characteristics of respondents and economics of main season rice and spring rice in Kanchanpur district are presented.

A. Socio-demographic Characteristics

The mean age of the MSR growers was 48.25 years where 49.33 years for SR growers. overall respondents of main season rice in the study area were 48.25 years and 11.19 years respectively. The gender distribution of main season and spring rice farmers, male showed the higher percentage in both the categories which is 63.33% and 66.67% respectively. In the main season rice area, the highest education level reported by respondents was predominantly "illiterate," constituting 31.7%. Conversely, in the spring rice cultivation area, the highest education level reported by respondents was primarily "secondary education," at 26.7%. This indicated a higher proportion of individuals who had completed at least secondary school. This trend reflected a comparatively more educated demographic engaged in spring rice cultivation. The dominance of secondary education in the spring rice area indicated a demographic with comparatively better access to education. This could have had implications for their engagement in rice cultivation practices, as higher education levels might have influenced decision-making, adoption of innovative techniques, and engagement with broader economic activities beyond agriculture.

The observation that the highest proportion of main season rice growers reported an income level in the range of 2 to 4 lakhs, which was 25 (30.15%), indicated a prevailing trend toward moderate incomes within this group as shown in Table I. This might be due to a mix of subsistence farming and additional revenue sources. In contrast, most spring rice growers reported an income of 6 lakhs and above (54.95%), suggesting a relatively affluent economic status, possibly influenced by factors like larger land holdings, more profitable farming practices, or diversification into higher-value crops.

TABLE I
SOCIO-DEMOGRAPHIC CHARACTERISTICS OF RICE GROWERS

Variable	Description	MSR	SR	Total
Age (years)	Average	48.25	49.33	48.79
Gender	Male	63.33	66.67	65.00
	Female	36.67	33.33	35.00
Education	Illiterate	19.00	12.00	15.50
	Primary	12.00	9.00	10.50
	Secondary	16.00	16.00	16.00
	University	6.00	9.00	7.50
Income	Average annual income (Rs/Yr)	553,333.33	606,666.67	580,000.00 ±182,236.34

B. Land Holding and Land Use

Table II compares key agricultural variables between Main Season Rice (MSR) and Spring Rice (SR) cultivation areas, including landholding, cultivated land, and irrigated land. While MSR exhibited a slightly higher mean landholding (0.65 ha) compared to SR (0.55 ha), the difference of 0.1 ha was not statistically significant (t-value = 0.520). In terms of cultivated land, MSR reported a significantly higher mean of 0.64 ha compared to SR's 0.39 ha, with a significant difference of 0.259 ha (t-value = 1.094). However, irrigated land showed marginal variation, with MSR at 0.60 ha and SR at 0.39 ha, and the 0.214 ha difference was not statistically significant (t-value = 0.928). Overall, these findings highlight a substantial difference in cultivated land while revealing minimal discrepancies in landholding and irrigated land between the two cultivation areas.

TABLE II
LAND HOLDING, CULTIVATED LAND AND IRRIGATED LAND UNDER MAIN SEASON AND SPRING RICE IN THE STUDY AREA

Variables in hectare	Main Season Rice (n=60)	Spring Rice (n=60)	Overall (n=120)	Mean difference	t-value
Land holding	0.65	0.55	0.6	0.10	0.520
Cultivated land	0.64	0.39	0.51	0.25	1.094*
Irrigated Land	0.60	0.39	0.49	0.21	0.928

Note: * indicates level of significance at 5%.

C. Cost of Production Analysis

1) *Cost of Production*: Table III compared the average production cost of rice in main season and spring rice farms. Spring rice had a higher average cost per hectare (NRs. 80,435.92) compared to main season rice (NRs. 75,199.84). This aligned with similar findings in the Rapti municipality, Chitwan district (Jaishi *et al.*, 2020), where spring rice production costs were above the national average (NRs. 64,967.72). Main season rice cultivation incurred a lower variable cost (NRs. 74,284.60) than spring rice (NRs. 78,924.91) per hectare. Labor use was higher in spring rice (48.10 man-days/ha) than main season rice (45.90 man-days/ha). Main season rice also used more seeds (57.25 kg/ha) compared to spring rice (45.01 kg/ha).

Notably, both main season and spring rice farmers did not incur insurance costs, suggesting a potential area for intervention or awareness campaigns to make crop insurance more accessible to these farmers.

TABLE III
AVERAGE COST OF PRODUCTION OF MAIN SEASON AND SPRING RICE IN THE STUDY AREA

Farm Operations	Unit	Main Season Rice (n=60)			Spring Rice (n=60)			Total (n=120)	
		Quant ity/ha	Total Cost (Rs/ha)	%	Quanti ty/ha	Total Cost (Rs/ha)	%	Quant ity/ha	Total Cost (Rs/ha)
Variable Cost									
Human labor	Man days	40.01	23756.00	31.59	41.49	23757.21	29.42	40.75	23756.61
Machine labor	Hour	5.89	16508.26	21.95	6.61	18116.62	22.52	6.25	17477.44
Total Labor		45.90	40264.27	53.54	48.10	41873.83	52.05	42.75	41069.05
Seed	Kg	57.25	3454.23	4.59	45.01	1462.84	1.82	51.13	2458.535
Manure and compost	Kg	3996.1	7992.15	10.62	4716.71	9433.42	11.73	4356.9	8712.73
Urea	Kg	101.01	3030.59	4.03	123.57	3707.33	4.61	12.2	3368.96
DAP	Kg	91.82	4591.24	6.11	66.83	3341.76	4.15	79.3	3966.5
Potash	Kg	0.53	1060.09	0.05	33.64	2979.91	3.70	1708	1511.195
Zinc	Kg	5.84	876.98	1.16	2.27	341.01	0.42	4.05	608.99
Agrochemicals	NRs.	1.42	284.01	0.37	2.35	471.23	0.59	1.88	142.94
Irrigation	NRs.	24.63	3695.27	4.92	19.86	2979.91	3.70	22.24	3337.59
Interest on working capital			9619.64	12.79		11961.07	14.87		10790.35
Miscellaneous Cost	NRs.		433.70	0.57		660.88	0.82		547.29
Total Variable Cost	NRs.		74,284.60	98.78		78,924.91	98.12		76,604.75
Fixed Cost									
Land Tax	NRs.		347.08	0.50		584.51	0.72		465.79
Water Tax	NRs.		58.05	0.07		143.66	0.17		100.85
Depreciation	NRs.		125.97	1.67		125.97	0.16		125.97
Repair and maintenance	NRs.		358.70	0.47		656.85	0.82		507.77
Total Fixed Cost	NRs.		889.82	1.22		1511.01	1.88		1200.41
Total Cost	NRs.		75,199.84	100.0		80,435.92	100.0		77,805.16

2) *Gross Return*: Table IV presented distinctive economic patterns in the gross return of rice cultivation, encompassing grain and straw, for main season rice (MSR) and spring rice (SR) cultivation areas. MSR had an average grain return of NRs. 88,147.53, constituting 85.74% of total returns. In Gorkha, Poudel *et al.*, (2021) found higher total returns for main season rice (NRs. 182,612). SR yielded NRs. 110,696.90, contributing 87.34% of the total. Jaishi *et al.*, (2020) also found similar results in the study performed in the Chitwan district for spring rice production. The overall mean returns from rice grain were NRs. 99,422.22, representing 86.62% of the total, with a statistically significant negative mean difference of NRs. -22,549.37 (t-value = -3.004). This indicates that, on average, MSR farmers generated lower rice grain returns than their SR counterparts.

TABLE IV
GROSS RETURNS FROM RICE PRODUCTION IN THE STUDY AREA

Returns (NRs. /ha)	Main Season Rice (n=60)	Spring Rice (n=60)	Overall (n=120)	Mean difference	t-value
Rice Grain	88,147.53 (85.74)	110,696.90 (87.34)	99,422.22 (86.62)	-22,549.37	-3.004*
Straw	14,665.57 (14.26)	16,038.92 (12.66)	15,352.25 (13.38)	-1,373.35	-2.513*
Total	102,813.10 (100.00)	126,735.82 (100.00)	114,774.47 (100.00)	-23,922.72	-5.517*

Note: * indicates level of significance at 5%. Figure in the parenthesis indicates percent.

Similarly, in terms of straw returns, MSR reported an average return of NRs. 14,665.57, constituting 14.26% of the total returns, while SR achieved slightly higher returns of NRs. 16,038.92, contributing 12.66% of the total. The overall mean straw returns were NRs. 15,352.25, representing 13.38% of the total, with a statistically significant negative mean difference of NRs. -1,373.35 (t-value = -2.513), illustrating that MSR farmers generated lower straw returns than SR farmers.

These findings collectively signified that MSR farmers, on average, yielded lower total returns compared to their SR counterparts, emphasizing significant economic differences between the two cultivation areas. The differences highlighted potential disparities in cultivation practices, market conditions, or resource management between MSR and SR.

3) *Productivity, Gross Margin, Gross Profit and Benefit-Cost Ratio*: Based on the results presented in Table V, there was a statistically significant difference in the yield of main season rice and spring rice. Specifically, MSR had a lower yield, with an average difference of 0.92 metric tons per hectare compared to SR. The t-value of 2.117 indicated that this difference was statistically significant at a 5% significance level. The average productivity for the MSR and SR was found to be 3.69 t/ha and 4.61 t/ha, respectively. This finding of main season rice was slightly higher than the finding of Sapkota *et al.*, (2021) in Rautahat district in Nepal.

In terms of economic performance, the comparison between MSR and SR cultivation areas revealed notable differences in both gross margin and gross profit. SR exhibited higher averages for both metrics: a gross margin of NRs. 47,810.96 compared to MSR's NRs. 28,503.08 and a gross profit of NRs. 46,299.95 compared to MSR's NRs. 27,613.26 (Table V). These discrepancies were statistically significant, with t-values of -2.483 and -2.387, respectively. Collectively, these findings indicated that, on average, SR farmers achieved higher gross margins and gross profits compared to MSR farmers, highlighting distinctive economic outcomes between the two cultivation areas. The higher gross margins and profits observed in SR suggested potentially more favourable economic conditions or strategies for rice cultivation in that area. Further analysis could delve into the factors contributing to these differences, such as input costs, yield variations, or market dynamics.

In MSR cultivation, the average BC ratio was 1.38, while SR achieved a higher average of 1.57, as presented in Table V. This finding of spring rice was higher than the finding in Chitwan district of Nepal, which was 1.21 Jaishi *et al.*, (2020). The result of main season rice was also found to be higher than the study conducted in Gorkha district for rice production (1.17) by Poudel *et al.*, (2021). The overall mean BC ratio for both areas was 1.47. This revealed that investment in spring rice production was economically viable in the study area. The observed negative mean difference of -0.19 indicated that, on average, MSR farmers experienced lower BC ratios compared to SR counterparts. This difference was statistically significant (t-value = -3.092), highlighting distinct economic variations between the cultivation areas. Differences in BC ratio suggested Spring Rice might bring better economic returns than Main Season Rice. This prompted a closer look at factors like costs, yields,

and markets. The significant negative mean difference and t-value highlighted a clear economic gap between the two areas.

TABLE V
PRODUCTIVITY, GROSS MARGIN, GROSS PROFIT AND BC RATIO OF RICE PRODUCTION IN THE STUDY AREA

Variable	Main Season Rice (n=60)	Spring Rice (n=60)	Overall (n=120)	Mean difference	t-value
Yield (t/ha)	3.69	4.61	4.15	-0.92	2.117*
Gross margin (NRs. /ha)	28503.08	47810.96	38157.02	-19307.88	-2.483*
Gross profit (NRs. /ha)	27613.26	46299.95	36956.61	-18686.69	-2.387*
BC ratio	1.38	1.57	1.47	-0.19	-3.092*

Note: * indicates level of significance at 5%.

4) *Production Function Analysis of Main Season and Spring Rice*: The regression analysis presented in Table VI provided valuable insights into the economic dynamics of main season and spring rice cultivation. For main season rice, the estimated F value (15.430) was statistically significant at 5%, indicating all the independent variables included in the model had good explanatory power to explain the variation in rice production. Notably, variables such as chemical fertilizer cost, agrochemical cost, and labor cost exhibited statistically significant positive impacts on returns, signifying their pivotal role in influencing economic outcomes. Subedi *et al.*, (2020) also reported the significant positive contribution of chemical fertilizer, human labor, and pesticide on the production of main season rice in Jhapa district. Conversely, seed cost and tillage cost did not display statistically significant effects. The reliability of the model was reinforced by a high adjusted R-squared value of 0.595, indicating that approximately 59.5% of the variability in returns could be attributed to the included variables.

The regression analysis of spring rice provided in Table VI delved into the impact of various cost factors on the economic performance of the studied variable. Among these factors, agrochemical cost, labor cost, and tillage cost emerged as statistically significant contributors, as indicated by their substantial positive coefficients, high t-values, and low p-values. This suggested that changes in these costs were associated with notable variations in the studied variable. Conversely, seed cost, manure cost, and chemical fertilizer cost lacked statistical significance in explaining variability, as their coefficients possessed relatively smaller magnitudes, and their t-values and p-values suggested limited influence. The model's robustness was underscored by a high adjusted R-squared value of 0.913, signifying the model's reliability in explaining the variations. With an F-value of 104.733, the overall significance of the model was established, and the presence of returns to scale (1.242) accentuated the potential amplification of returns with increased inputs.

In comparing the regression results between main season rice and spring rice, notable differences emerged. The R-squared value for main season rice stood at 0.636, explaining approximately 63.6% of the yield variation, like findings by Dhakal *et al.*, (2019) in the Chitwan district. In contrast, the R-squared for spring rice was higher at 0.757, indicating that the model accounted for approximately 75.7% of the variation in spring rice yield, suggesting a better fit for spring rice data. The Returns to Scale value, exceeding 1 for spring rice, implied increasing returns to scale, indicating that scaling up resources was more effective for spring rice production compared to main season rice. In summary, these findings underscored distinct relationships between input costs and yield for the two rice types, with spring rice displaying a stronger association and more pronounced scalability, although the significance of these relationships varied across specific input costs for each crop.

TABLE VI
COBB-DOUGLAS PRODUCTION FUNCTION ANALYSIS OF MAIN SEASON AND SPRING RICE PRODUCTION IN THE STUDY AREA

Variables	Main Season Rice			Spring Rice		
	Coefficient	t-value	p-value	Coefficient	t-value	p-value
Seed cost	-0.072	-1.205	0.234	0.100	0.966	0.338
Manure cost	0.029	1.677	0.100	0.021	1.126	0.265
Chemical Fertilizer cost	0.141**	3.068	0.003	0.081	0.926	0.359
Agrochemical cost	0.034**	2.657	0.010	0.355***	4.614	0.001

Labor cost	0.750***	8.194	0.001	0.151*	2.186	0.033
Tillage cost	-0.141*	-2.277	0.027	0.534***	4.594	0.001
Constant	3.906***	3.775	0.001	3.433***	4.394	0.001
F-value	15.430			27.545		
R square	0.636			0.757		
Adjusted R-square	0.595			0.730		
Returns to Scale	0.741			1.242		

Notes: *, ** and *** indicate level of significance at 5%, 1% and 0.1% respectively.

5) *Returns to Scale*: The sum of coefficients of independent variables in the main season rice model was 0.74, indicating decreasing returns to scale, while for spring rice, it was 1.24, reflecting increasing returns to scale (Table VI). Like this, Poudel *et al.*, (2021) reported a decreasing return to scale in main season rice production in Gorkha with a value of 0.56. Comparing returns to scale values between main season rice and spring rice cultivation areas provided insights into their production dynamics. The returns to scale value of 0.74 for MSR suggested a balanced and efficient scale of production, where an increase in inputs led to a proportional rise in returns. In contrast, the higher returns to scale value of 1.242 for SR indicated that the spring rice cultivation area benefited from an even more pronounced increase in returns as inputs expanded. This suggested that SR cultivation had a greater potential for scaling up production with increased inputs, implying a conducive environment for achieving higher yields and economic gains. This comparison underscored the differential scalability of the two cultivation areas, revealing that SR cultivation might offer more favourable conditions for optimizing returns through input expansion.

6) *Production Constraints for Main Season and Spring Rice Production in the Study Area*: There were many constraints obtained by the farmers on paddy production. This study attempted to find some of them as recognized by the farmers. Similarly, the five-point scaling technique was used to find the relative intensity or priority of the constraints. The major production constraints reported by the farmers during the main season and spring rice cultivation were ranked in Table VII. In the case of MSR, top production constraints included unavailability of quality inputs on time, diseases, insect pests' infestation, lack of modern irrigation technology, and limited access to improved technology and harvesting equipment. For SR farmers, the major constraints were unavailability of quality inputs on time, diseases, insect pests' infestation, poor technical knowledge about improved farming practices, and lack of modern irrigation and processing technology.

The most severe constraint for both main season rice ($I=0.85$) and spring rice growers ($I=0.88$) was the unavailability of quality inputs on time, followed by diseases and insect pests' infestation (Table VII). Jaishi *et al.*, (2020) used the scaling technique to identify constraints associated with spring rice in eastern Chitwan, Nepal, highlighting challenges in fertilizer supply from neighbouring India during peak hours.

TABLE VII
PRODUCTION CONSTRAINTS FOR MAIN SEASON AND SPRING RICE PRODUCTION IN THE STUDY AREA

Production Problems	Main Season Rice		Spring Rice	
	Index	Rank	Index	Rank
Unavailability/ limited availability of improved technology including equipment	0.57	IV	0.49	VI
Unavailability of quality inputs on time	0.85	I	0.88	I
Diseases and insect pests' infestation	0.79	II	0.71	II
Inadequate modern technology in irrigation	0.58	III	0.50	IV
Poor technical knowledge about improved farming practices	0.49	VI	0.59	III
Insufficient modern technology in harvesting	0.55	V	0.47	VII
Abiotic stresses like flood	0.28	IX	0.35	IX
Unavailability of modern technology in processing	0.41	VIII	0.50	V
Inadequate modern technology in threshing and drying	0.46	VII	0.45	VIII

7) *Satisfaction with Production and Profitability of Rice Crops*: In the main season rice cultivation area, satisfaction distribution was balanced, with 51.7% holding a neutral stance, 18.3% expressing satisfaction, and 30% indicating dissatisfaction or high dissatisfaction (Table VIII). In the spring rice cultivation area, 43% were

neutral, while 53.4% expressed satisfaction, including 11.7% highly satisfied, and only 3.3% reported dissatisfaction. Spring rice respondents generally held a more positive sentiment toward rice crop production and profitability, suggesting a greater perceived success rate compared to the main season rice area, which exhibited a more balanced distribution with a significant neutral stance and relatively more dissatisfied respondents.

TABLE VIII
SATISFACTION WITH PRODUCTION AND PROFITABILITY OF RICE CROPS AMONG MAIN SEASON AND SPRING RICE RESPONDENTS

Level of satisfaction	Main Season Rice Frequency	Spring Rice Frequency	Total Frequency
Highly Satisfied	0 (0.00)	7 (11.70)	7 (5.85)
Satisfied	11 (18.30)	25 (41.70)	36 (30.00)
Neutral	31 (51.70)	26 (43.30)	57 (47.50)
Dissatisfied	15 (25.00)	2 (3.30)	17 (14.15)
Highly Dissatisfied	3 (5.00)	0 (0.00)	3 (2.5)
Grand Total	60 (100.00)	60 (100.00)	120

Note: Figure in the parenthesis indicates percent.

D. Technology Adoption

1) *Involvement in Cooperative Organization*: The analysis of involvement in cooperative organizations across main season rice and spring rice revealed key trends (Table IX). Farmers groups held a significant presence, with 29 (48.33%) of MSR respondents and 34 (56.67%) of SR respondents participating. Cooperatives also exhibited consistent engagement, with 15 (25.00%) of respondents in both areas being involved.

TABLE IX
INVOLVEMENT OF MAIN SEASON AND SPRING RICE RESPONDENTS IN COOPERATIVE ORGANIZATIONS

Involvement in Cooperative Organizations	Main Season Rice Frequency	Spring Rice Frequency	Total Frequency
Cooperatives	15 (25.00)	15 (25.00)	30 (25.00)
Farmers group	29 (48.33)	34 (56.67)	63 (52.50)
Others (Banking Institutions, etc.)	16 (26.67)	11 (18.33)	27 (22.50)
Grand Total	60 (100.00)	60 (100.00)	120 (100.00)

Note: Figure in the parenthesis indicates percent.

The "others" category showed relatively higher engagement in MSR, 16 (26.67%), compared to SR, 11 (18.33%). This underscored the prevalent role of collaborative farming practices, as evidenced by the substantial involvement in farmers groups, and the balanced participation in cooperatives suggested their importance in both types of rice cultivation.

IV. CONCLUSIONS AND RECOMMENDATIONS

This study comprehensively analyzed socio-demographic, economic, and agricultural aspects in Kanchanpur district's main season rice (MSR) and spring rice (SR) cultivation areas. Demographic trends highlighted active involvement in rice cultivation, with disparities in gender and education. Farming strategies, land use, and cultivation practices differed significantly, with semicommercial farming more predominant in SR. Economic indicators revealed variations, with SR incurring higher costs but generating better outcomes. Spring rice cultivation stood out for efficient technology use, addressing labor shortage issues. The study provides valuable insights for policymakers and practitioners, emphasizing the need for context-specific interventions. Further research in different regions can contribute additional perspectives to enhance our understanding of rice cultivation in Nepal.

Based on the findings of this study, several suggestions can be drawn to enhance agricultural practices and economic outcomes in both Main Season Rice (MSR) and Spring Rice (SR) cultivation areas:

- Mechanization: Promote agricultural machinery adoption to reduce human labor costs. For promotion of spring rice, an appropriate harvester combined with drying facility is recommended for increased coverage, and production.
- Improved seed selection: Encourage the adoption of improved and locally adapted seed varieties to enhance production and reduce input costs, considering the variations in seed usage and costs between the two areas.
- Skill enhancement and training: Develop customized training programs to address unique needs in both cultivation areas. Prioritize training on innovative techniques like System of Rice Intensification (SRI) and Integrated Pest Management (IPM) to enhance farming practices.
- Future research: Further in-depth research into the variations in economic performance, production constraints, and profitability between MSR and SR is essential.

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